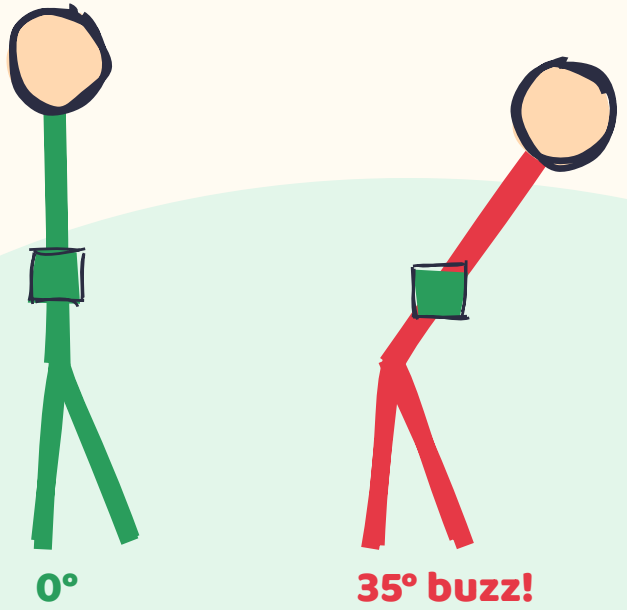
Young Innovators · Class 22

# Smart Posture Corrector Wearable

## A gentle buzz when you slouch

Neck load | Tilt angle | Smart alert



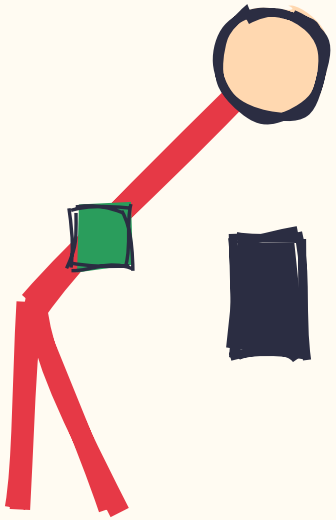

Did you know?
Looking down at a phone at 60° puts about 27 kg of force on your neck, the weight of an 8-year-old child!

Let's think...

# How heavy is your head right now?

**Phone neck**

Your head weighs about 5 kg. Hold it upright and your neck barely notices. Lean forward to look at your phone and the load on your neck **multiplies**. Hours a day, every day...



5 kg?

It's light!

It depends on angle!

**Today:** you build a **wearable** that measures your back angle and buzzes gently when you slouch for too long.

Learning objectives

# By the end of this class I can...

| | | |
|---|---|---|
| 1 | **Why posture matters:** explain how **leaning forward** multiplies the load on your neck and back | **Proof:** I can explain the '27 kg head' to a friend |
| 2 | **Measure tilt:** calculate a **tilt angle** from an MPU6050 accelerometer using **atan2** | **Proof:** Serial shows 0° upright and about 30° when I lean |
| 3 | **Smart alert:** buzz a **vibration motor** only after slouching for **5 seconds**, with calibration | **Proof:** my wearable buzzes when I slouch, not when I reach for a pen |

Objective 1 · Why posture matters · video

# Video time!




**The benefits of good posture**

TED-Ed · about 4 minutes · youtu.be/OyK0oE5rwFY

**Watch for...**

1. Which **muscles** work harder when we slouch?
2. What happens to the **discs** in our spine?
3. Which habits help good posture?

Scan to watch

**After the video:** Sit the way you are sitting now. Would the video say it's good posture? Fix one thing.

Objective 1 · Why posture matters

# Leaning multiplies the load

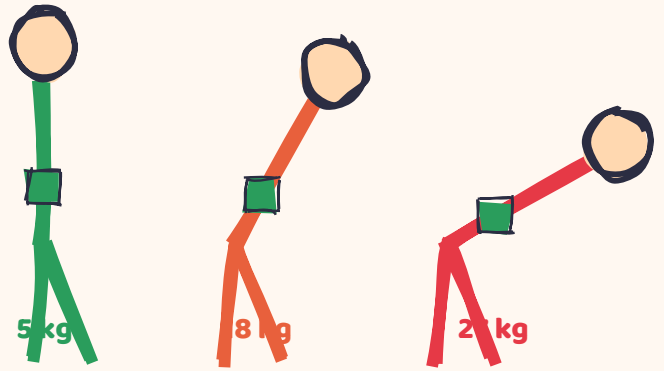
## The idea

Your head is a 5 kg weight on top of a stick (your spine). Upright, the weight sits **over** the spine. Lean forward and it acts like a **lever**: neck muscles must pull much harder.

Research (Hansraj, 2014): 0° ≈ 5 kg, 15° ≈ **12 kg**, 30° ≈ **18 kg**, 60° ≈ **27 kg**.

Fix: screen at eye level, back supported, **move** every 30 minutes.

## Example: holding a school bag




Hold your bag close to your chest: easy. Hold it out at arm's length: your arm shakes. Same bag, longer lever. Your head works the same way.

Objective 2 · Measure tilt

# An accelerometer feels gravity

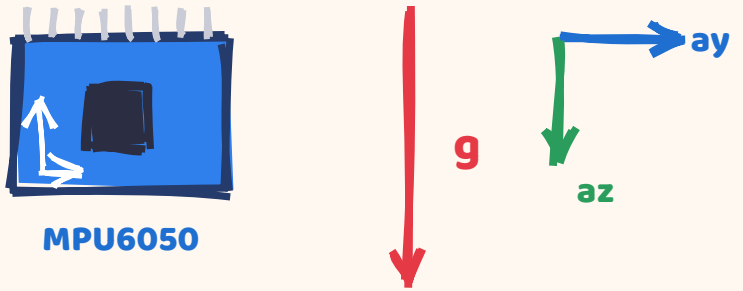
## The idea

The MPU6050 measures acceleration on **3 axes** (x, y, z). When still, the only acceleration is **gravity** (1 g, pointing down).

Upright: gravity is all on one axis. Tilt: gravity **shares** between two axes. So:
**angle = atan2(ay, az) × 180 / π**

atan2 turns the two parts back into an angle in degrees.

## Example: a glass of water




Tilt a glass and the water surface stays level while the glass leans. By comparing the water to the glass, you can tell the angle. Gravity is the sensor's 'water'.

Objective 3 · Smart alert

# Smart alerts don't nag

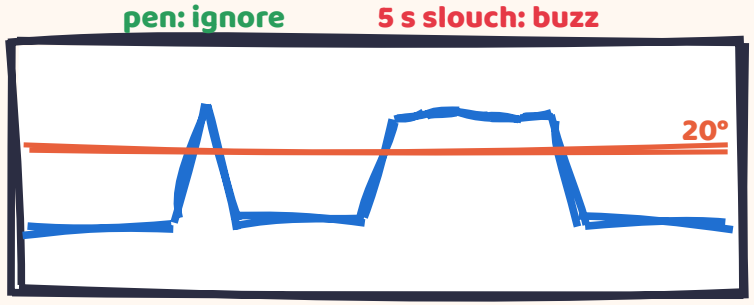
## The idea

A dumb alarm buzzes every time you lean to pick up a pen. People rip it off in a day!

Smart design:

1. **Calibrate**: at start, sit straight; save that angle as 'zero'.
2. **Threshold**: slouch = more than **20°** past zero.
3. **Time delay**: buzz only if slouching for **5 s** in a row.
4. **Gentle**: short buzz, not a siren.

## Example: a car seatbelt warning




The car doesn't beep the second you sit down. It waits a few seconds, then beeps. That delay is what makes it helpful instead of annoying.

Real world

# Wearables that coach



**Posture trainers**
Upright and Lumo sell clip-on buzzers like ours



**Fitness bands**
Remind you to move every hour



**Physiotherapy**
Sensors track patients' exercises at home



**Factory workers**
Wearables warn when lifting with a bent back



Would you wear this all day? What would make it comfortable and not embarrassing?

**Design for humans**
The best wearable is one people actually keep wearing. Comfort and calm alerts matter as much as accuracy.

Quick quiz

# Posture physics

| Question | Answer |
| --- | --- |
| 1. About how much load at 60° lean? | |
| 2. When still, what does an accelerometer measure? | |
| 3. Upright = 2°, now = 28°. Lean past zero? | |
| 4. Why wait 5 s before buzzing? | |
| 5. What does calibration save? | |

Answers

Cover! 1: ≈ 27 kg · 2: gravity · 3: 26° · 4: ignore quick moves like reaching · 5: your own 'straight' angle

Part 2 · Make it

# Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Human posture sensor
2. Our wearable kit
3. Read the tilt angle
4. Calibrate, threshold, delay, buzz
5. Wear it and tune it
6. Wearer test
7. Wearable designer challenges

Class game

# Human posture sensor

## How to play

1. Pairs: one is the **wearer**, one is the **sensor** with a protractor or phone level app.
2. Wearer sits straight: sensor says '0°'. Wearer slowly leans: sensor calls angles.
3. Rule: if over 20° for a count of 5, sensor taps their shoulder (the buzz!).
4. Wearer tries to pick up a pen fast. Should the sensor buzz?

## Then discuss

Where on the body is the best place for the sensor: neck, upper back, or lower back? Why?

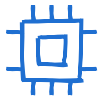


Big idea

Where you place a sensor changes what it measures.

Build it

# Our wearable kit



**Arduino Nano**
Small enough to wear



**MPU6050**
Accelerometer + gyro (I2C)



**Vibration motor module**
Gentle buzz



**9 V or USB power bank**
Portable power



**Elastic strap / clip**
Holds it on the upper back



**Button**
Press to calibrate

Wiring: MPU SDA → A4, SCL → A5, VCC → 5V. Vibration IN → D5. Button D2 → GND (INPUT_PULLUP).

Library
Install 'MPU6050' by Electronic Cats (or Adafruit MPU6050) from Library Manager.

Objective 2 · Measure tilt · Build step 1 of 3

# Read the tilt angle

```
#include <Wire.h>
#include <MPU6050.h>
MPU6050 mpu;

float tiltDeg() {
  int16_t ax, ay, az;
  mpu.getAcceleration(&ax, &ay, &az);
  return atan2(ay, az) * 180.0 / PI;     // forward/back tilt
}

void setup() {
  Serial.begin(9600);
  Wire.begin();
  mpu.initialize();
  Serial.println(mpu.testConnection() ? "MPU OK" : "MPU FAIL");
}

void loop() {
  Serial.println(tiltDeg());
  delay(200);
}
```

**Test**

Open Serial Plotter. Tilt the board forward and back. A smooth line?

**Wrong axis?**

If tilting changes nothing, try atan2(ax, az) instead: it depends how you mount it.

**MPU FAIL?**

Check SDA/SCL are not swapped and VCC is connected.

MAKE

Objective 3 · Smart alert · Build step 2 of 3

# Calibrate, threshold, delay, buzz

```ino
const int VIB = 5, BTN = 2;
float zero = 0;  unsigned long slouchStart = 0;

void calibrate() {
  float sum = 0;
  for (int i = 0; i < 20; i++) { sum += tiltDeg(); delay(50); }
  zero = sum / 20;                        // your 'straight'
  digitalWrite(VIB, HIGH); delay(100); digitalWrite(VIB, LOW);
}

// in setup(): pinMode(VIB, OUTPUT); pinMode(BTN, INPUT_PULLUP); calibrate();

void loop() {
  if (digitalRead(BTN) == LOW) calibrate();
  float lean = tiltDeg() - zero;
  if (abs(lean) > 20) {
    if (slouchStart == 0) slouchStart = millis();
    if (millis() - slouchStart > 5000) {   // 5 s in a row
      digitalWrite(VIB, HIGH); delay(300); digitalWrite(VIB, LOW);
      slouchStart = millis();              // wait 5 s before next buzz
    }
  } else {
    slouchStart = 0;                       // sat up: reset timer
  }
  delay(100);
}
```

**millis()**

Counts milliseconds since power on. We remember WHEN slouching began.

**Average**

20 readings averaged = a steadier 'zero'.

**Reset**

Sitting up sets slouchStart back to 0, so quick leans never buzz.

Objective 3 · Build step 3 of 3

# Wear it and tune it

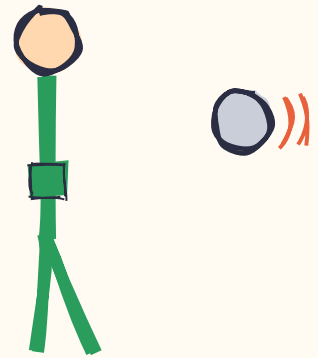
**a** Strap the board to the **upper back** (between shoulder blades) with the elastic.

**b** Sit straight, press the button: one short buzz = calibrated.

**c** Slouch for 5 seconds. Buzz? Reach for a pen quickly. No buzz?

**d** Tune: is 20° too strict or too loose for you? Try 15° and 25°.

**e** Wear it for 10 minutes of normal work. Count the buzzes.



**Check:** Slouching for 5 s buzzes; quick moves don't.

Never strap electronics tightly or under clothes against skin for long. Remove it if it gets warm.

Test like an engineer

# Wearer test

| Test | Result | Unit |
|---|---|---|
| Upright reading after calibration | | ° |
| Slouch reading | | ° |
| Quick pen reach: buzzed? | | |
| Buzzes in 10 min of work | | |
| Best threshold for me | | ° |

## Debug checklist

Angle jumps around? Average 5 readings in tiltDeg().
Always buzzing? Calibrate while sitting straight, check the sign of lean.
Never buzzing? Print lean to Serial and compare with 20.

Accelerometers get confused by movement (walking adds acceleration). That's why it works best when sitting.

MAKE

Level up!

# Wearable designer challenges

## Choose your challenge



**Gyro fusion:** combine accelerometer + gyro (complementary filter) for a steadier angle.



**Posture score:** count good vs slouch minutes and show a % on Serial.



**App:** use an ESP32 and send your posture score to your phone.



**Comfort:** design a 3D-printed clip that fits the electronics neatly.

### Think like an engineer

Why might the same threshold not work for every person?

**Boss level**

Make it fully wireless with a small LiPo battery and wear it for a whole lesson.

Objective check

# Can I really do it? Prove it.

| | | |
|---|---|---|
| 01 | Explain why looking down at a phone at 60° is hard on your neck. | ○ ○ ○ |
| 02 | Show Serial: what angle does your sensor read upright and slouched? | ○ ○ ○ |
| 03 | Demonstrate: slouch 5 s = buzz, quick reach = no buzz. | ○ ○ ○ |

**Rate yourself**

Colour one circle each. Red: not yet. Amber: with help. Green: I can explain it to a friend.

**Teacher:** Posture challenge: who can get through a 10-minute task with zero buzzes?

Show what you know

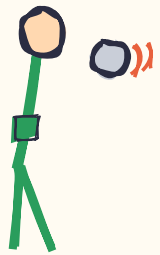
# What did we learn?

- [x] Leaning turns your head into a heavy lever
- [x] Accelerometers measure gravity when still
- [x] angle = atan2(ay, az) × 180 / π
- [x] Calibrate + threshold + delay = alerts people accept
- [x] millis() lets you time without delay()

**Exit ticket:** why don't we buzz the moment the angle passes 20°?

At home

Check your posture while doing homework tonight. Move your screen or book to eye level.



**Next class:** Li-Fi Messenger: Sending Text with Light!